

Utility Asset Registry

How to operate this system

Electricity distribution utility · Bhubaneswar

Simple command guide for different users

20 September 2026

This booklet is for people who run the system day to day. You do not need to know Python. Read only the section for your job: IT (Section A), night operator (Section B), day staff / surveyor (Section C), administrator (Section D), or supervisor (Section E).

Project folder: C:\Users\ranja\utility-asset-registry

This system stores electricity assets in Bhubaneswar (poles, transformers, valves, manholes). Field staff send a CSV file. The system cleans it, saves good rows, and keeps bad rows for correction.

You do **not** need to know Python. You type short words in PowerShell.

Project folder: C:\Users\ranja\utility-asset-registry

Read only the section for your job:

If you are	Read
IT / first-time installer	Section A
Night operator (load today's file)	Section B
Day staff / surveyor	Section C
Administrator	Section D
Supervisor (check quality)	Section E

Everyone should also read **Commands** and **If something goes wrong**.

Open PowerShell (every user, every time)

- 1 Press the **Windows** key.
- 2 Type **PowerShell**.
- 3 Press **Enter**.
- 4 Type these two lines, then press **Enter** after each:

```
cd C:\Users\ranja\utility-asset-registry
.\.env\Scripts\Activate.ps1
```

What this means: go to the project folder, then turn on the project tools. You must do this **once in each new PowerShell window** before load, start, check, or the other words will work.

If PowerShell asks about running scripts, type:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Then try `.\.env\Scripts\Activate.ps1` again.

Commands (one word each)

Type this	Who uses it	What it does
setup	IT, first time only	Installs the tools and creates the password file <code>.env</code>
load	Night operator	Reads <code>data\survey_export.csv</code> , saves good rows, writes bad rows to a rejects file
start	Day staff, admin	Starts the system. Keep that window open. Then open <code>http://127.0.0.1:8000/docs</code>
check	Anyone	Tells you if the system is running
rejects	Night operator, supervisor	Opens the bad-rows file from the last load
run	Anyone who wants a menu	Shows: 1 = start, 2 = load, 3 = exit

If PowerShell treats `start` as a Windows command instead of this system, type:

```
asset start
```

asset load, asset check, asset rejects, asset setup, and asset run work the same way.

A. IT / first-time installer

Your job: prepare the computer once. Other people then use load and start every day.

1. Install Python

Install **Python 3.11 or newer** if it is not already installed. During install, tick **Add Python to PATH**.

2. Open PowerShell and go to the folder

```
cd C:\Users\ranja\utility-asset-registry
```

3. Create the tool folder and install once

```
python -m venv .env  
.\.env\Scripts\Activate.ps1  
pip install -e .  
setup
```

Line	Meaning
python -m venv .env	Makes a private folder of tools for this project
Activate.ps1	Turns those tools on
pip install -e .	Installs this system so load and start work
setup	Installs packages and creates .env if it is missing

4. Set the first administrator password

```
notepad .env
```

Change these three lines, then **Save** and close Notepad:

- JWT_SECRET= a long secret sentence that only IT knows
- BOOTSTRAP_ADMIN_USERNAME=admin (or another username)
- BOOTSTRAP_ADMIN_PASSWORD= the password staff will use to sign in the first time

Do not share this file. Do not put a real password in any file that is sent to GitHub.

5. Prove it works

```
start
```

Keep that window open. In a **second** PowerShell window (same cd and Activate steps), type:

```
check
```

You should see: **The system is running.**

Open a browser: **http://127.0.0.1:8000/docs**

To stop the system, click the first window and press **Ctrl+C**.

B. Night operator

Your job: at the end of the shift, load today's handheld GPS file. You do **not** need the browser for this.

1. Put today's file in the right place

Copy the CSV to:

```
C:\Users\ranja\utility-asset-registry\data\survey_export.csv
```

If a file is already there, replace it.

The file must have these column names:

```
asset_id, name, asset_type, latitude, longitude, elevation_m, surveyed_on, surveyor, status,
condition_score, attribute_json
```

2. Open PowerShell (see the top of this document)

3. Load the file

```
load
```

Meaning: read the CSV, clean each row, save good assets, write bad rows to outputs\rejects.csv.

4. Read the result on the screen

Example:

```
Rows read:      62
Rows accepted:  52
Rows rejected:  10
Rejects file:   outputs\rejects.csv
```

Line	Meaning
Rows read	How many rows were in the file
Rows accepted	Good rows saved in the system
Rows rejected	Bad rows not saved
Rejects file	Where the bad rows were written

Good rows are already saved. Bad rows are not thrown away.

5. If rejected rows are more than 0

```
rejects
```

Meaning: opens the bad-rows file (usually in Excel). Look at the last column named reason. Tell your supervisor. After the file is corrected, put it back as data\survey_export.csv and type load again.

Optional: menu instead of load

```
run
```

Then press 2 and Enter.

C. Day staff / surveyor

Your job: search assets, add a missing one, or correct a field. You cannot delete assets, create users, or upload a CSV in the browser. That is for an administrator.

1. Start the system

Open PowerShell (see the top of this document), then:

```
start
```

Meaning: turns the system on. **Keep this window open.** If you close it, the browser will stop working.

If you are not sure it is on, open a second PowerShell window and type check.

2. Open the work page

In Chrome or Edge open:

http://127.0.0.1:8000/docs

This page lists every action. It is not a map. The map is a separate program that talks to this system.

3. Sign in

- 1 Open **POST /auth/login**.
- 2 Click **Try it out**.
- 3 Put your username and password:

```
{"username": "your-username", "password": "your-password"}
```
- 1 Click **Execute**.
- 2 Copy the **token** from the response.
- 3 Click **Authorize** at the top of the page.
- 4 Paste: Bearer then a space then the token.
- 5 Click **Authorize**, then **Close**.

Meaning: the system now knows who you are. The token expires after about 60 minutes. Sign in again if work stops.

4. Daily work (after you are signed in)

What you want	Where on the page
List assets	GET /assets
Find one asset by code	GET /assets/PL-0001 (use the real code, like PL-0142)
Search by name or text	GET /assets and fill in q
Filter by type, status, or surveyor	GET /assets and fill in those boxes
Add an asset	POST /assets
Correct some fields	PATCH /assets/{code}
Replace all fields	PUT /assets/{code}
See visit history	GET /assets/{code}/visits
Counts for the map	GET /reports/summary
Assets that need repair	GET /reports/repairs
Nearest asset to a point	GET /reports/nearest

Asset codes look like PL-0142: two letters, a hyphen, four digits.

Allowed types: pole, valve, manhole, transformer. Allowed status: active, decommissioned, proposed. Condition score: **0 to 10**. A decommissioned asset cannot have a score above 2.

5. Stop the system

Click the PowerShell window that is running start, then press **Ctrl+C**.

D. Administrator

Your job: everything a surveyor can do, plus delete assets, create users, and upload a CSV in the browser.

1. Start and sign in

Same as day staff: type start, open **http://127.0.0.1:8000/docs**, sign in with the **admin** username and password.

The first admin account comes from .env (BOOTSTRAP_ADMIN_USERNAME and BOOTSTRAP_ADMIN_PASSWORD). Change that password with IT. Do not leave change-me-before-go-live in use.

2. Extra actions only an admin may do

What you want	Where on the page
Create a surveyor or another admin	POST /auth/users
Delete an asset (and its visit history)	DELETE /assets/{code}
Upload a day's CSV from the browser	POST /ingest/upload

When you create a user, choose role surveyor or admin. Give the person the username and password yourself. This system does not send email.

If a surveyor tries to delete or upload, the system answers: **Only an administrator may do that.**

3. Upload in the browser vs night load

- Night operator load is the usual way: file goes in data\survey_export.csv.
- **POST** /ingest/upload is the same cleaning, used when an admin already has the CSV and the system is running.

Both save good rows and report rejected rows.

E. Supervisor

Your job: see whether today's load was clean, and see which assets need repair.

You can use commands (no browser) or the reports page (browser).

After the night load (no browser)

- 1 Ask the night operator for the counts (read, accepted, rejected).
- 2 If rejected is more than 0, type rejects (after the usual PowerShell steps).
- 3 Open reason in Excel. Send the bad rows back to the field team.
- 4 After they fix the file, the night operator types load again.

Also look in the outputs folder:

File	What it is
outputs\rejects.csv	Bad rows plus a reason
outputs\summary.txt	Short printable summary
outputs\assets.geojson	Good assets for a map
outputs\ingest.log	One dated line per load

In the browser (after someone types start and you sign in)

What you want	Where
Totals and map extent	GET /reports/summary
In-service assets with condition below 5	GET /reports/repairs
Assets visited most often	GET /reports/most-visited
Who surveyed on a date	GET /reports/surveyors

If something goes wrong

What you see	Meaning	What to do
Put the CSV file here first	data\survey_export.csv is missing	Copy today's file to that path, then load
Please run setup / command not found	Tools are not installed, or you skipped Activate	IT: run Section A. You: run the two lines at the top
CSV is missing required column(s)	Header names are wrong	Fix the CSV column names, then load
The system is not running	start is not open	Type start in a window and leave it open
Username or password is wrong	Sign-in failed	Check .env with IT, or ask admin for your account
Sign in is required	No token, or token expired	Sign in again and click Authorize
Only an administrator may do that	Surveyor tried admin work	Ask an admin
Too many requests / 429	You clicked too fast	Wait one minute and try again
start opens a new Windows window	PowerShell used the Windows start word	Type asset start instead

What you should never do

- Do not close the **start** window while people are using the browser.
- Do not delete **.env** or share the admin password.
- Do not change **run.bat** for developer tests. Night staff may still use the menu (**run** or **run.bat**).
- Do not send **dev.db** or **.env** to people outside the team.

Quick cards (print these)

Night operator

```
cd C:\Users\ranja\utility-asset-registry
.\.venv\Scripts\Activate.ps1
load
```

If rejected rows > 0, type **rejects**.

Day staff

```
cd C:\Users\ranja\utility-asset-registry
.\.venv\Scripts\Activate.ps1
start
```

Keep the window open. Browser: <http://127.0.0.1:8000/docs> Sign in → Authorize → do your work. Stop with **Ctrl+C**.